When Should an LLM Search? A Calibration-Centric Evaluation of Search Behavior and a Hidden Generalization Gap Under Natural Paraphrases

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Abstract

Modern LLMs deployed as info-seeking assistants must decide, on the fly, whether to lean on their parameters or reach out to a live web search. As both parametric knowledge and agentic tool use grow, the right interplay between recall and retrieval becomes genuinely unclear — yet existing evaluations measure end-task accuracy and question-level tool use, never asking the upstream question: *given that a model can search, does it search when it should, and skip when it shouldn't?* We propose a protocol that pairs the model's own internal knowledge boundary (per-hop uncertainty, estimated by re-sampling) with its revealed search decisions (per-hop attribution of each emitted query), and apply it to three production-style models from independent vendors. We find that search calls are barely aligned with the model's own uncertainty, that the issued queries are overwhelmingly spent off-target, and that a cosmetic paraphrase — one that preserves the information need but drops the benchmark's explicit hop chain — causes every model to search less *and* re-allocate the searches it retains *away from* the hops it is uncertain about. The dominant failure mode flips from wasted retrieval (a cost the user pays in latency) to uncovered gaps (a cost the user pays in correctness) — a hidden generalization gap that existing benchmarks do not measure.

1 Introduction

Info-seeking queries — requests where a user asks the model to find, verify, or assemble facts — are the canonical setting in which modern LLMs are paired with web-search tools, and retrieval-augmented LLMs are now a default deployment pattern for them (Huang and Huang, 2024). Two simultaneous trends make the right interplay between parametric recall and live retrieval on these queries increasingly opaque. First, frontier models keep gaining parametric knowledge, so for any given fact the chance that the answer is already inside the model is non-negligible and growing. Second, the same models have grown markedly more agentically capable, so emitting additional search calls is essentially free at the policy level — but not free in latency, dollars, or the risk that retrieved context overrides a correct parametric belief (Wu et al., 2024a; Tao et al., 2024). There is no consensus, theoretical or empirical, on what the *right* division of labor between parametric memory and live search should look like in this regime; in particular, neither "always search" nor "search only when the closed-book answer is wrong" is operationally well-defined when the model is a black box with no exposed knowledge boundary.

The standard evaluation question is whether retrieval improves end-task scores; a far less studied question is whether the model uses retrieval *appropriately* on these info-seeking queries — i.e. whether it triggers searches on facts it does not know and skips them on facts it does. Tool-use benchmarks such as WTU-Eval (Ning et al., 2024) frame the problem at the granularity of the entire question (use the tool or not), and the broader RAG / tug-of-war literature (Cheng et al., 2024a; Wu et al., 2024a; Tao et al., 2024; Kim et al., 2025; Farahani and Johansson, 2024) focuses on how models *integrate* external evidence once retrieved. Per-step retrieval gating has been studied in pipeline-RAG settings (Jiang et al., 2023; Asai et al., 2024; Yao et al., 2025; Guan et al., 2025; Wu et al., 2025a), and recent work shows that even these binary-gate decisions are brittle to surface-form variation (Jang et al., 2026). Our work extends this analysis to a qualitatively different setting: a multi-hop *agentic* info-seeking system with live web-search tool access, where the agent must decide *what* to query, *when*, and whether to continue, uncertainty propagates across reasoning hops, and the evaluation is grounded in per-hop parametric uncertainty rather than a corpus-retrieval gate or outcome correctness.

This is the question we address. Crucially, we deliberately do *not* ground our analysis in end-task correctness. The premise of the paper is that an external retrieval decision should be aligned with the model's own internal epistemic state — regardless of whether the model would, downstream, happen to produce the right final answer. A model that searches questions it knows perfectly and skips questions it doesn't is *mis-calibrated* as a search agent, independently of whether retrieval rescues the final answer.

Research questions. The paper is organised around four questions that the protocol is built to answer, in the chronological order in which we asked them. Each results section opens by naming the question it addresses and closes with a one-line verdict.

RQ1. Are models' search calls aligned with their epistemic uncertainty? (§4)

RQ2. Are models' search calls efficient? (§5)

RQ3. Do models decompose questions and search efficiently? (§6)

RQ4. Do models have different behaviour on benchmark questions vs. real users' questions? (§7)

We define *search efficiency* concretely as the fraction of issued queries that target a hop the model is actually uncertain about — the Search Budget Efficiency (SBE) of §5; equivalently, one minus the wasted-search rate. Throughout the paper we also reflect, at the end of each RQ section, on whether the observed behaviour is good — and synthesise those reflections in the Discussion (§8).

Contributions.

1. An **evaluation protocol** that operationalises the parametric↔search interplay on agentic info-seeking queries: per-hop semantic entropy (the model's own internal knowledge boundary) paired with per-hop search attribution (the model's revealed retrieval decision) on multi-hop questions (§3).
2. A **six-category behavioural taxonomy** (three correct cells, three distinct error cells) that makes the interplay legible at hop granularity, including the *Cross-hop covered* cell that strict "did the model search this hop" framings mis-classify as a miss (§6).
3. A **hidden generalization gap**: a cosmetic, meaning-preserving paraphrase shifts all three models toward *Truly missed* as the dominant error cell, with the gap traceable to specific reasoning hops and to a behavioral mechanism (early per-hop commitment) we localise in §8.

2 Related Work

Tool-use and search-decision evaluation. WTU-Eval (Ning et al., 2024) measures tool-use at the *question level*. Step-level retrieval gating has been studied extensively in *pipeline-RAG* settings: FLARE (Jiang et al., 2023) triggers retrieval on low-probability tokens; Self-RAG (Asai et al., 2024) learns inline reflection tokens; DeepRAG (Guan et al., 2025) formalises per-subquery retrieval as an MDP; HiPRAG (Wu et al., 2025a) assigns hierarchical process rewards per reasoning step; ReaLM-Retrieve (Guo et al., 2026) introduces a per-step Reasoning-Step Uncertainty Signal tested on MuSiQue. SMART-ER (Qian et al., 2025) and Wu et al. (2025b) define over-search and under-search rates as explicit calibration metrics. Toolformer (Schick et al., 2023) and RARE (Wang et al., 2025b) train search insertion during generation but evaluate downstream task accuracy rather than per-decision calibration. All of this work operates over fixed document-retrieval pipelines; the search decision is a binary switch, not a free-form query to a live web-search tool inside an iterative reasoning loop. Our work evaluates this qualitatively different, harder setting — agentic info-seeking with a live web tool — and grounds calibration in per-hop parametric uncertainty rather than outcome correctness.

Parametric vs. contextual knowledge — the interplay that motivates this paper. A growing line studies how LLMs balance internal priors against retrieved context (Cheng et al., 2024a; Wu et al., 2024a; Farahani and Johansson, 2024; Kim et al., 2025; Wadhwa et al., 2026; Tao et al., 2024; Zhang et al., 2024). These works probe *conflict resolution* — what happens when internal and external knowledge disagree — rather than the upstream question of *when to invoke retrieval at all* on an info-seeking query. More recent work has moved toward per-token conflict arbitration: AdaCAD (Wang et al., 2025a) measures per-step Jensen-Shannon divergence between parametric and contextual distributions; CoCoA (Khandelwal et al., 2025) uses a composite per-step signal (Rényi divergence, entropy gap, contextual peakedness) to gate which logits dominate; Micro-Act (Huo et al., 2025) forces an explicit intermediate self-verification step before committing to a retrieved fact. We treat all of these as direct motivation: collectively they show the parametric↔contextual interplay is fragile even *after* retrieval has been triggered — context can override correct parametric knowledge at the token level. The natural consequence is that on info-seeking queries the *upstream* decision — whether to retrieve at all, given that the model may already have the answer parametrically — is itself a load-bearing calibration problem, not a free action. As frontier models continue to gain parametric knowledge in parallel with stronger agentic tool use, the desired shape of this upstream policy becomes increasingly unclear, and yet has remained largely unmeasured at hop granularity in the agentic setting.

Uncertainty quantification. We use semantic entropy (Farquhar et al., 2024; Kunitomo-Jacquin et al., 2026) as our operational definition of "knows / does not know." Semantic entropy correlates well with hallucination (Farquhar et al., 2024) and is cheaper than fine-grained atomic factuality (Min et al., 2023); Semantic Entropy Probes (Kossen et al., 2024) further reduce cost to a single forward pass via hidden-state linear probes. SEARAG (SEARAG, 2025) applies per-hop semantic entropy gating inside a pipeline-RAG multi-hop setting — the closest prior work in spirit to our per-hop uncertainty measurement; we extend this to the agentic setting and use entropy as a *post-hoc diagnostic* rather than a training-time gate.

Multi-hop QA and compositional reasoning. MuSiQue (Trivedi et al., 2022) supplies the hop-decomposed questions we need; MINTQA (He et al., 2024) probes the boundary between parametric memorization and genuine retrieval-dependent reasoning with gold sub-question annotations. The compositional-brittleness literature (Li et al., 2024; Yang et al., 2024; Wu et al., 2024b) documents how multi-hop performance collapses with hop-count and under surface-form variation: CREME (Li et al., 2024) shows that a repaired model must survive semantically equivalent paraphrases of the same two-hop chain. We add: search *behavior* in agentic systems is brittle to surface form independently of reasoning accuracy. For trace-level evaluation methodology we follow Lee and Hockenmaier (2025); the validity of LLM-judge per-step attribution is supported by Han et al. (2025), who find that models ≥70B reach or exceed human-to-human inter-rater agreement on step-level annotation tasks.

Framing and paraphrase sensitivity. Framing sensitivity in retrieval-pipeline systems has recently been documented: Jang et al. (2026) measure retrieval-decision flip rates of up to 55% under meaning-preserving paraphrases in adaptive RAG; Cheng et al. (2026) formalise the "knowing-doing gap" — a mismatch between a model's internal recognition that a tool is needed and its actual generation of tool-call tokens, amplified under conversational framing; Rabinovich et al. (2025) document output-mode collapse (structured tool-call tokens dissolving into prose) under prompt rephrasing in function-calling settings. Our work extends this line to a multi-hop *agentic* context: by coupling surface-form variation with per-hop uncertainty tracking we identify *which specific reasoning hops* are abandoned under natural framing — a diagnostic that pipeline-level analyses cannot provide.

Action calibration. Standard calibration asks whether expressed confidence correlates with answer correctness (Farquhar et al., 2024). A complementary notion — *action calibration* — asks whether the decision to invoke a tool is aligned with the model's actual epistemic state. Wang et al. (2025c) formalise this as the Decision-Knowledge Alignment Principle: tool-use decisions are miscalibrated whenever the decision boundary diverges from the internal knowledge boundary. TARG (Wang et al., 2026) and SUGAR (Zubkova et al., 2025) calibrate retrieval gates using logit margins and semantic entropy respectively; SeaKR (Yao et al.,

2025) uses hidden-state Gram determinants. Our protocol is a *diagnostic* complement to these training-time and inference-time gating methods: rather than imposing a threshold, we measure post-hoc how well search decisions of three production models align with their parametric uncertainty at hop granularity.

3 Protocol

We evaluate a model M on a set of multi-hop questions $Q = \{q_i\}$, each decomposing into an ordered chain of hops $h_{i,1}, \ldots, h_{i,k_i}$ with gold answers $a_{i,j}$. The protocol assigns every hop a label drawn from a six-cell behavioural taxonomy (§6) by combining a per-hop parametric-uncertainty probe with an analysis of the model's revealed search decisions on the aggregate question. Figure 1 illustrates the procedure schematically; the formal pseudocode is deferred to Appendix E.1.

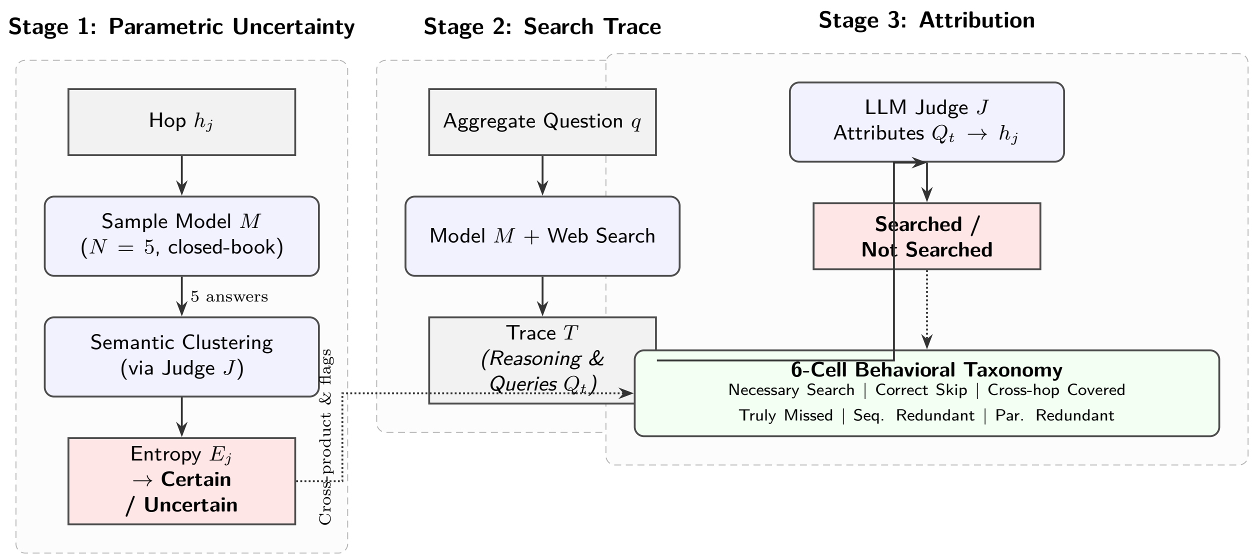

Figure 1: **Protocol schematic.** *Left:* For each hop h_j, the model is sampled N=5 times without search; semantic clustering yields the entropy E_j and a CERTAIN/UNCERTAIN label. *Middle:* The same model answers the aggregate question with web search enabled, producing a trace of interleaved reasoning and search calls. *Right:* An LLM judge attributes each emitted query back to a specific hop; the cross-product of certain × searched, refined with sequential-redundancy and cross-hop-coverage flags, yields the six-cell behavioural taxonomy used throughout the paper.

Stage 0 — staleness filtering. MuSiQue contains time-sensitive questions (office-holders, populations) whose correct action depends on external state rather than internal knowledge. An LLM judge flags such items; we retain **600 non-stale multi-hop questions** as the evaluation set.

Stage 1 — per-hop parametric uncertainty. We follow Farquhar et al. (2024) and estimate uncertainty from N=5 closed-book samples clustered into semantic equivalence classes, computing semantic entropy E_j in bits. We use sampling-based rather than logit-based entropy because the Gemini API does not expose per-token log-probabilities and because RLHF-sharpened token distributions systematically under-estimate genuine ignorance (Jiang et al., 2023; Farquhar et al., 2024). A hop is **certain** iff $E_j = 0$ *and* the cluster matches the gold answer — the conjunction guards against the confident-wrong failure mode.

Stage 2 — search-augmented run. The model receives the aggregate question and a Brave-Search-backed web tool; we record the full trace and the final answer. Brave operates an independent web index with no corporate product relationship to the three evaluated model vendors, reducing provider-side confounds. We do not fine-tune any model.

Stage 3 — query-to-hop attribution and trace flags. An LLM judge maps each emitted search query to a single hop using the hop questions and the preceding reasoning as context; a secondary multi-hop pass validates this simplification (79–87% of queries serve a single hop; Appendix E.6). Two trace-level flags enrich the binary searched$_{i,j}$: `seq_redundant` marks hops whose search was sequentially redundant within the trace, and `xhop_cov` marks hops whose gold answer was returned as a by-product of a query attributed to a different hop.

Six-cell behavioural taxonomy. Cross-tabulating these annotations partitions every per-hop decision into one of six interpretable categories (Table 1). Three are correct decisions; three are search errors with distinct semantics. We will refer to these cells throughout the paper.

Table 1: The six search-calibration categories. The first three are correct decisions; the remaining three are distinct error modes. "Searched" refers to whether the LLM-judge attributed at least one search query to the hop. `seq_red` = sequentially redundant within the trace; `x-hop` = the gold-equivalent content was returned by a query attributed to a different hop.

Category	**Certain?**	**Searched?**	`seq_red`	`x-hop`	**Reading**
Necessary search	no	yes	–	–	Correct: search was warranted
Correct skip	yes	no	–	–	Correct: model knew, did not waste a call
Cross-hop covered	no	no	–	yes	Correct: hop recovered via another hop's query
Truly missed	no	no	–	no	Error: gap left open
Seq. redundant	yes	yes	yes	–	Error: re-querying already-known content
Par. redundant	yes	yes	no	–	Error: searching a hop the model knew

Models. We evaluate **Gemini 3.1 Pro** (Google), **Qwen 3.5 122B** (Alibaba), and **Nemotron Nano 30B** (NVIDIA) — three vendors, three architectural families, three independent training pipelines, spanning roughly one order of magnitude of scale, all with native tool-use support. Behavioural patterns shared across all three are therefore unlikely to reflect a common training artifact. Operational details (serving, judge choice, search backend, paraphrase generation) are deferred wholesale to Appendix E.

4 RQ1: Are Searches Aligned with Epistemic Uncertainty?

RQ1 asks whether a model searches a hop *because* it is uncertain about that hop — the most direct test of whether the search policy is governed by the model's own knowledge boundary. We test alignment with the AUROC of entropy → search: under an uncertainty-driven policy this would approach 1.0. We observe instead AUROC in the 0.56–0.62 range across all three models (Figure 2).

Two observations sharpen the finding. (1) Even the best of the three, Nemotron, clears AUROC 0.6 only modestly — and pays for that small calibration advantage with the highest *Truly missed* rate (anticipated in §6), so what alignment exists is purchased by coverage failure rather than by graded sensitivity. (2) The per-entropy search rate is roughly flat rather than monotone in E_j for all three models: search behaves as an approximately binary action with little graded response to the magnitude of internal uncertainty (Appendix B).

Verdict (RQ1). No — search calls are barely aligned with the model's own epistemic uncertainty. The latent uncertainty signal exists in the model and is measurable, but it is not the dominant driver of when any of these three production-style LLMs decide to search.

Is this good behaviour? This is the most fundamental miscalibration we will see: a search policy that ignores the model's own knowledge boundary cannot be good in either regime. But whether the resulting errors land as costs (wasted retrieval) or as correctness failures (uncovered gaps) depends on *what specifically* gets mis-searched — the question we turn to next in §5 and §6.

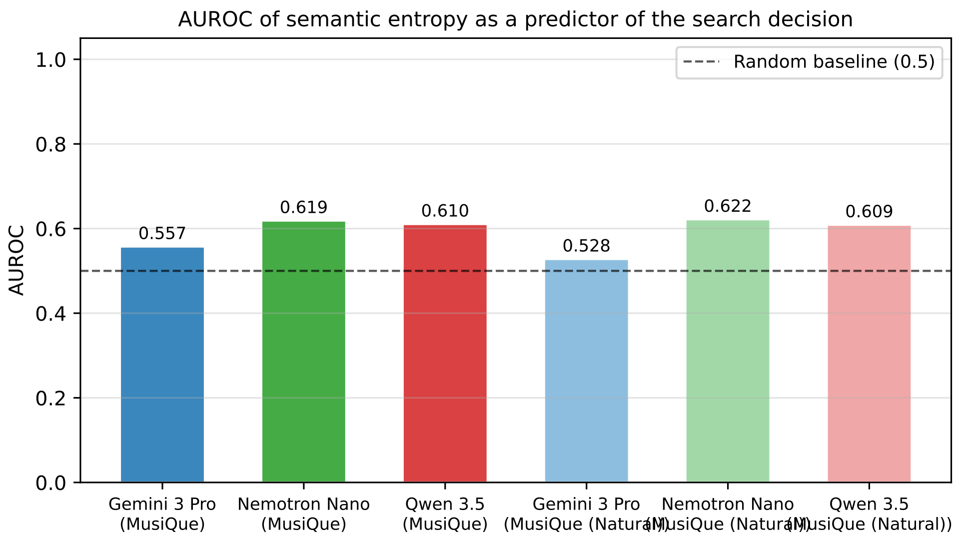

Figure 2: AUROC of semantic entropy as a predictor of the per-hop search decision, per (model, dataset). The dashed line marks the random baseline (AUROC = 0.5). All six combinations fall in the 0.53–0.62 range; for Gemini under natural paraphrasing, AUROC drops to 0.528 — effectively chance.

5 RQ2: Are Search Calls Efficient?

If a model's searches are not driven by its own uncertainty (§4), how much of its query budget actually lands on the hops where it is needed? **RQ2** asks this directly, formalised as the *Search Budget Efficiency* (SBE) — the fraction of issued queries that target an uncertain hop, equivalently one minus the wasted-search rate:

$$\text{SBE} \;=\; \frac{N_{\text{queries on uncertain hops}}}{N_{\text{queries total}}}.$$

SBE collapses both redundancy modes (parametric, sequential) into a single "budget spent off-target" rate. On the benchmark, Gemini scores SBE = 0.045, Qwen 0.232, and Nemotron 0.233 (Figure 3): even the best of the three spends roughly 77% of its query budget off-target, and Gemini spends 95%.

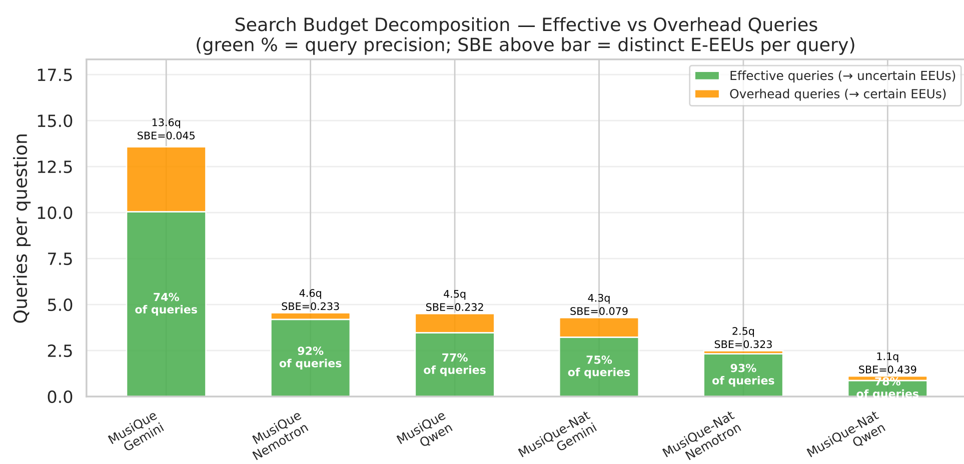

Figure 3: Search Budget Efficiency per model and per dataset. SBE is the fraction of issued queries attributed to a hop with $E_j > 0$ (an uncertain hop); the remainder is wasted on parametrically- or sequentially-redundant searches. Higher is better.

Verdict (RQ2). No — between $\sim 77\%$ (Nemotron, Qwen) and $\sim 95\%$ (Gemini) of every model's query budget is spent off-target relative to its own knowledge boundary.

Is this good behaviour? An SBE of 0.045 is not a defensible operating point: the model is paying for an order-of-magnitude more queries than it consumes information from. Even Nemotron's 0.233 means three-quarters of its budget is wasted. But the scalar number does not tell us *how* each model is inefficient — whether the wasted spend is double-checking facts already known (a cost) or leaving uncertain hops uncovered (a correctness failure). That is RQ3.

6 RQ3: Do Models Decompose Questions and Search Efficiently?

RQ3 asks not just whether the search budget is efficient on aggregate (RQ2) but *how* each model's inefficiency is distributed across the six taxonomy cells of Table 1 — whether a model's failure is one of wasted searches or of uncovered gaps. Decomposition itself is not in serious doubt: the LLM judge attributes 79–87% of queries to a single hop, confirming that all three models do syntactically split the question into per-hop searches. What differs sharply is the qualitative shape of the inefficiency.

Table 2 gives the full six-category breakdown per model on the benchmark. The extended categories distinguish two flavours of redundancy (sequential vs. parametric) and isolate the Cross-hop covered category, which a strict "did the model search this hop" framing would otherwise mis-classify as a miss.

Table 2: Search-calibration profile per model on MuSiQue (% of 1800 hops). Left of the rule: correct decisions. Right: distinct error modes.

Model	Correct			Errors		
	Necessary search	**Correct skip**	**Cross-hop covered**	**Truly missed**	**Seq. redundant**	**Par. redundant**
Gemini 3.1 Pro	44.7	15.6	5.1	5.1	9.2	**20.5**
Nemotron Nano 30B	43.3	15.4	10.2	**24.3**	1.3	5.4
Qwen 3.5 122B	**49.0**	13.9	7.7	6.7	2.8	19.8

Three model archetypes. The category mix reveals qualitatively different calibration failures:

- **Gemini** is a *high-volume, redundancy-prone* searcher. Its dominant error mode is *Parametrically redundant* (20.5%) plus *Sequentially redundant* (9.2%) — nearly 30% of its hops are wasted searches on content it already has, either in its parameters or in earlier turns of the trace. Coverage of uncertain hops, however, is excellent: *Truly missed* is only 5.1%.
- **Nemotron** is the inverse archetype — a *coverage-failing under-searcher*. Redundancy of both kinds is small (1.3% + 5.4%), but *Truly missed* dominates at 24.3%: nearly a quarter of all hops are uncertain and the model never even attempts a query.
- **Qwen** sits between the two. It searches more uncertain hops than the other two (49.0% *Necessary search*) but pays for it with 19.8% *Par. redundant*, mirroring Gemini's redundancy pattern at slightly lower volume.

A Kruskal–Wallis test on the per-hop entropy distributions rejects the null of equal distributions across the three models ($H = 499.2$, $p < 10^{-108}$), so the behavioural differences are not driven by trivially different uncertainty inputs; full statistics are in `statistical_tests.csv` (Appendix E.2).[1]

The three-way error decomposition is essential. Any binary collapse to "correct search-decision" versus "error" would hide that Gemini and Nemotron fail in opposite ways. Cross-hop coverage — the case where an uncertain hop is recovered as a by-product of another hop's query — is non-negligible (5.1–10.2%) for all three; we count it as a correct outcome, since the hop's information need is in fact satisfied, but it is invisible in a strict "did the model search this hop" framing and would be misread as a miss without this annotation.

[1] A Mann–Whitney U test on the entropy of *searched* hops further shows Nemotron searches substantially more uncertain hops on average than Gemini ($p < 10^{-97}$), consistent with the two models' very different positions on the redundancy/coverage axis.

Verdict (RQ3). Yes on decomposition; no on efficiency. All three models split the question and dispatch hop-level queries, but the resulting per-hop search profile is dominated by distinct error modes: redundancy for Gemini and Qwen, missed coverage for Nemotron.

Is this good behaviour? The three archetypes are not equally bad. Gemini's redundancy spend wastes latency and dollars but rarely loses correctness; Nemotron's coverage gaps directly cost answer correctness; Qwen sits between the two. Which is preferable depends on whether the deployment context tolerates latency or hallucination — a tension we return to in §8.

7 RQ4: Do Models Behave Differently on User-Style Questions?

RQ4 asks whether the behavioural profile we have just measured transfers to questions phrased the way a real user would phrase them. MuSiQue exposes the hop chain syntactically ("What administrative territorial entity includes the place where X was educated?"). A genuine user with the same information need would phrase the question differently ("I'm curious about X's background — what region of the country did he study in?"). The information need is identical (same gold answer, same hop chain), but the surface form differs: no nested relative clauses, no exposed bridge entities, often a request for a short narrative rather than a single fact. If the search-decision policy were governed by the model's internal uncertainty about facts, behaviour should be approximately invariant to this surface change. We test this directly by rewriting each of the 600 MuSiQue questions into a natural, goal-oriented paraphrase under three verifiable constraints (no bridge entities named, no exposed relative-clause hop structure, phrasing that invites 2–4 explanatory sentences; Appendix E.7) and re-running only Stages 2–3.

Observation 1 — Search volume collapses. All three models cut search volume substantially (Table 3): Qwen by 4.2×, Gemini by 3.1×, Nemotron by 1.8×. The information need is identical; only the phrasing changed.

Table 3: Search volume per question, benchmark vs. natural paraphrase (paired across 600 items). Wilcoxon signed-rank test, two-sided.

Model	**Benchmark (mean)**	**Natural (mean)**	**Ratio**	**Wilcoxon** p
Gemini 3.1 Pro	15.20	4.93	3.1×	3.0×10^{-73}
Nemotron Nano 30B	5.74	3.13	1.8×	2.4×10^{-21}
Qwen 3.5 122B	5.60	1.33	4.2×	6.4×10^{-88}

Observation 2 — Calibration collapses: the wrong cells grow. Reduced search would be unsurprising if the model's calibration *survived*: a leaner search budget could still be allocated to uncertain hops. It does not. Table 4 shows the six-category shift from benchmark to natural paraphrase for each model. The pattern is uniform across models: *Necessary search* collapses by 11–27 percentage points, and the lost mass moves predominantly into *Truly missed* (uncertain hop the model fails to search at all). *Par. redundant* also drops — which would be a positive in isolation — but the trade-off is unfavourable: the model loses correct searches faster than it loses wasted ones.

Table 4: Six-category profile shift under natural paraphrasing (Δ = Natural − Benchmark, in percentage points of 1800 hops). All three models lose *Necessary search* and gain *Truly missed.*

	Correct			**Errors**		
Model	ΔNec. srch.	ΔCorr. skip	ΔX-hop cov.	ΔTruly miss.	ΔSeq. red.	ΔPar. red.
Gemini 3.1 Pro	−19.4	+14.3	+2.0	+**17.3**	−6.1	−8.2
Nemotron Nano 30B	−10.5	+3.4	+1.4	+**9.2**	−0.7	−2.7
Qwen 3.5 122B	−27.4	+15.5	−0.3	+**27.7**	−2.4	−13.1

Observation 3 — The calibration gap persists or widens. Under natural paraphrasing the AUROC of entropy → search drops marginally for Gemini (0.557 → 0.528) and is essentially unchanged for the two open models. RQ1's finding — that entropy is a weak predictor of search — *persists and degrades for the highest-volume searcher.* Surface form, not internal uncertainty, governs the policy.

The headline picture. Figure 4 compresses the shift into a single chart: the four-cell (E / PR / M / CP) collapse of the six-category profile per model and per dataset, with each (model, dataset) bar summing to 100% of its 1800 hops. Reading left-to-right, the first three bars are the three models on benchmark MuSiQue and the next three are the same models on the natural paraphrase. Across all three models the red *Missed* segment expands sharply, the green *Effective* segment contracts, and the orange *Parametrically Redundant* segment also shrinks — but the shrinkage of PR is not a saving, because the same models lose Effective search faster than they lose wasted search.

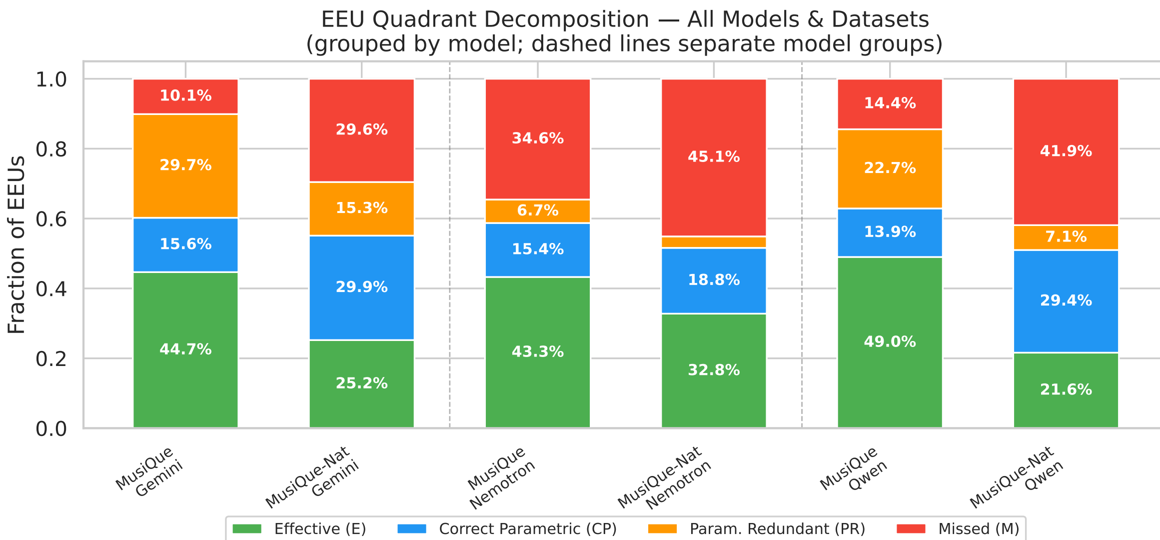

Figure 4: **Headline figure for RQ4.** Four-cell taxonomy collapse of the six-category profile (E = Effective / Necessary search, CP = Correct skip, PR = Parametrically Redundant, M = Missed) per model and per dataset. Each bar sums to 100% of that (model, dataset)'s 1800 hops. The natural-paraphrase bars show a uniform expansion of red (Missed) and contraction of green (Effective).

The per-model extended profile of Figure 5 (Gemini; remaining two models in Appendix A) confirms the cell-level direction of the shift. A complementary appendix probe (Appendix D) shows that the growth in *Truly missed* is dominated by *genuine* misses rather than by downstream cascade failures from a wrong earlier hop, so the shift is not an artifact of upstream errors.

Verdict (RQ4). Yes — behaviour changes sharply under a cosmetic paraphrase that preserves the information need, and the change is not benign. All three models search less, the remaining searches are re-allocated *away* from uncertain hops, and the dominant new error cell is *Truly missed*: an uncertain hop the model never queries the web for. The model's internal knowledge boundary has not moved; only the surface form of the question has.

Is this good behaviour? No. The shift is not benign: under benchmark framing the error budget concentrates in *Parametrically Redundant* (PR — wasted but recoverable; the user pays in latency); under natural framing it concentrates in *Truly missed* (M — an answer-correctness failure). The natural regime trades cost-flavoured errors for correctness-flavoured errors — a strictly worse exchange from the user's perspective. This re-allocation is the load-bearing piece of the Discussion.

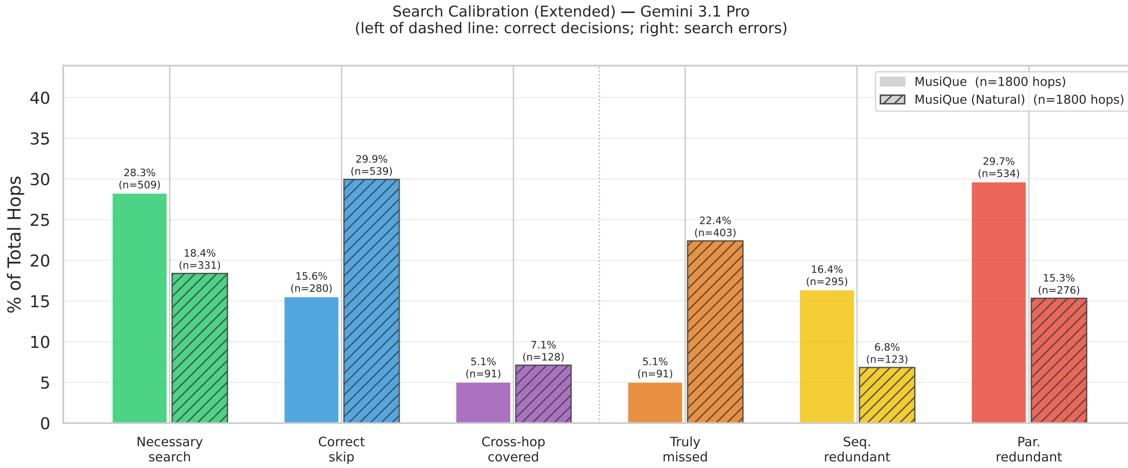

Figure 5: Extended search-calibration profile for Gemini 3.1 Pro. Solid bars: benchmark MuSiQue; hatched bars: natural paraphrase. Left of the dashed line are correct decisions; right are the three error categories. The benchmark→natural shift moves mass from Necessary search into Truly missed (and from Par. redundant into Correct skip), the calibration regression Table 4 quantifies. The other two models are in Figure 6.

8 Discussion

Synthesis: which behaviour is better, and in what cases? The four results sections each closed with a reflection on whether the observed behaviour is good; here we pull those reflections together. The benchmark regime is the more searched-on, more redundancy-prone of the two: every model's error budget under benchmark framing concentrates in *Parametrically* and *Sequentially Redundant* searches — queries the model did not need to issue, paid for in latency and dollars but rarely costing correctness. Under natural framing the error budget migrates into *Truly missed* (Table 4: +17, +9, +28 percentage points for Gemini, Nemotron, Qwen), a strictly worse currency: an uncertain hop never queried and never covered, so the answer either hallucinates or omits the information need. Across deployment contexts the choice is not symmetric. In latency-sensitive applications where wrong answers can be tolerated downstream, the benchmark regime's redundancy is the smaller harm; in correctness-sensitive applications where a confident wrong answer is the worst possible failure, even the benchmark regime is not safe and the natural regime is unacceptable. The same re-allocation argument applies across models: Gemini's redundancy is a cost, Nemotron's missed coverage is a failure, Qwen's mix is the intermediate trade-off — and the cost-vs-failure axis, not raw search volume, is the right ordering principle.

Why framing flips the policy — the commitment-locus mechanism. A behavioural probe localises *when* in the trace the regression happens (full results in Appendix C). For every (example, hop, variant) pair an LLM judge identifies the first turn at which the agent commits to a per-hop answer. The decisive cell is `commit_no_search`: the agent settled on an answer for that hop and no search was ever attributed to it. Under benchmark framing, Gemini exhibits this pattern on 22.6% of hops; under natural framing the rate jumps to 51.9%. Qwen rises 18.0% → 38.5%. The model is not "forgetting" to search — it is *deciding* the answer earlier, before any search query is issued. This is a behavioral analogue of the "context leads, parametric memory follows" effect of Tao et al. (2024): here the *search-decision* policy, not the answer, conditions on the prompt's surface register, consistent with the mechanistic account of Cheng et al. (2026) in which a model's tool-recognition and tool-execution representations become nearly orthogonal under conversational framing.

Implications for tool-use training. The standard RAG / tool-use training paradigm — Toolformer-style supervised insertion (Schick et al., 2023), Self-RAG critique tokens (Asai et al., 2024), distillation from agentic trajectories (Wang et al., 2025b) — does not explicitly tie the decision to search to the model's own uncertainty about the sub-step. Methods that close this loop — e.g. S-GRPO (Suri et al., 2025), which rewards search only when the model's belief warrants it — exist but are not yet the dominant paradigm. Our protocol offers a diagnostic complement to these training-time fixes: it measures, on a held-out set, whether a given training method actually closes the calibration gap on both benchmark and user-style framings.

Limitations. (i) Per-hop attribution uses an LLM judge; the one-query–one-hop simplification is validated in Appendix E.6. (ii) Sampling-based entropy at $N{=}5$ gives a coarse seven-value uncertainty resolution; we do not probe logits because the Gemini API does not expose them. Richer probes (Kunitomo-Jacquin et al., 2026; Kossen et al., 2024) may sharpen the signal. (iii) MuSiQue is the only multi-hop benchmark we evaluate, because it is the only one that provides the gold hop decomposition our Stage 1 protocol needs; generalisation to other multi-hop datasets is left to future work. (iv) We do not vary the search backend; results may differ with stronger retrieval. (v) Only one paraphrase per question is evaluated; the generalization gap may be larger across the full natural-language manifold.

9 Conclusion

Info-seeking queries put parametric memory and live web search in continuous tension. Existing benchmarks tell us whether RAG helps on aggregate; they do not tell us whether a model knows *when* to defer to its parameters and when to reach for the web. The protocol introduced here asks this question directly, grounded in per-hop semantic entropy paired with per-hop search attribution. Applied to three production-style LLMs (Gemini 3.1 Pro, Qwen 3.5 122B, Nemotron Nano 30B), it produces the following one-line verdicts:

- **RQ1.** No — search calls are barely aligned with internal uncertainty; the AUROC of entropy $\rightarrow$ search sits in 0.53–0.62, barely above chance, and search is approximately binary in internal uncertainty.
- **RQ2.** No — Search Budget Efficiency ranges from 0.045 (Gemini) to 0.233 (Nemotron), so between 77% and 95% of every model's query budget is spent off-target.
- **RQ3.** Yes on decomposition; no on efficiency. All three models split the question (79–87% one-to-one query-to-hop attribution) but exhibit qualitatively different inefficiency profiles: redundancy-prone (Gemini), coverage-failing (Nemotron), or in between (Qwen).
- **RQ4.** Yes — a cosmetic, meaning-preserving paraphrase causes every model to behave very differently. Search volume drops 1.8–4.2$\times$ and the remaining searches are re-allocated *away* from the same uncertain hops the model identified on the benchmark, with *Truly missed* becoming the dominant error cell.

Together these constitute a hidden generalization gap in the parametric$\leftrightarrow$search interplay of contemporary info-seeking LLMs. As both axes of model capability — parametric knowledge and agentic tool use — continue to advance, the protocol introduced here is a starting point for measuring, and ultimately closing, that gap.

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A Extended Search-Calibration Plots for Nemotron and Qwen

Per-model extended-calibration plots for the two models not shown in §6. The bars use the same colour code as Figure 5: green / blue / purple on the left of the dashed line are correct decisions; orange / yellow / red on the right are the three distinct error categories.

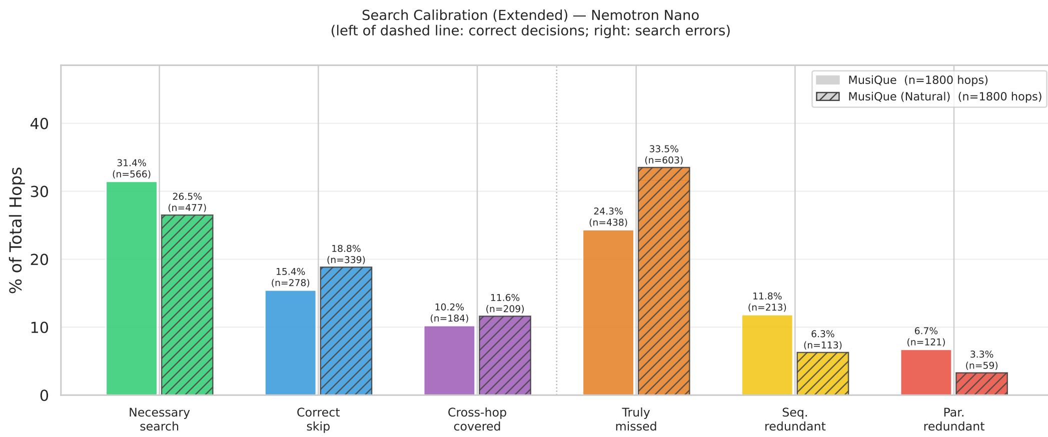
B Binary Search Behavior across Entropy Levels

A direct calibration view plots the search rate $P(\text{searched} \mid H = h)$ as a function of the discrete entropy level h produced by $N = 5$ closed-book sampling. Under well-calibrated behavior, this curve would rise monotonically from H=0 (the model should rarely search; it knows the answer) to H=2.32 (the model should almost always search; it has near-uniform disagreement across samples). What we observe instead (Figure 7) is a roughly flat curve for all three models: the search rate at H=0 is only a little below the rate at the highest entropy levels, and the intermediate levels are noisy. A χ^2 test of uniformity across entropy levels does reject the null for every model ($p < 10^{-4}$), so a small signal is statistically present — but the practical shape is binary: the model either searches at a roughly-constant rate or it doesn't, with little graded sensitivity to how uncertain it is. This is the per-entropy view of the near-chance AUROC reported in §4.

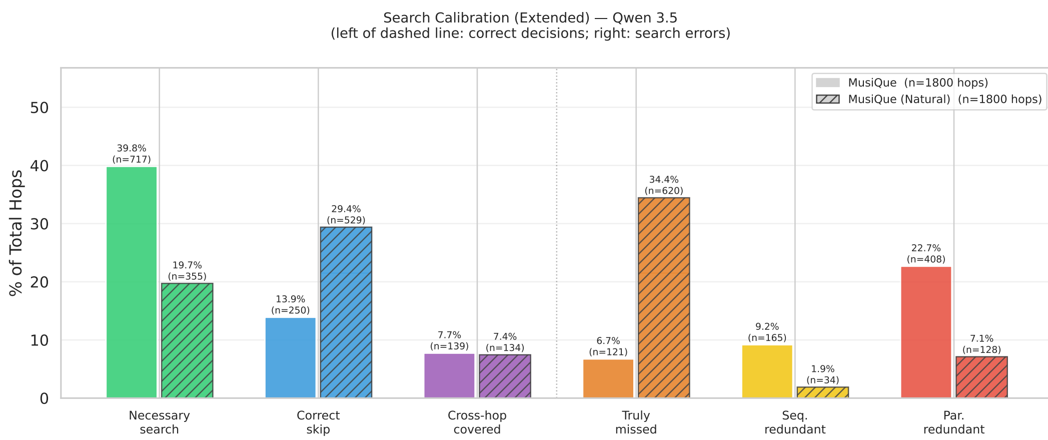
C Commitment-Locus Probe

Motivation. If natural framing causes models to skip searches they would otherwise have issued, *when* does the skip happen? One hypothesis is that the casual register of the paraphrased question invites a quick narrative answer, leading the model to commit to a per-hop answer early — before any search query is issued for that hop. The *commitment-locus probe* tests this directly.

Probe. For every (example, hop, variant) appearing in both the benchmark trace and the natural trace of the same model, an LLM judge (`gpt-oss:120b`) identifies the first turn at which the agent commits to a specific per-hop answer (asserts "X is Y" or treats Y as established and uses it downstream). Combined with the first turn at which a search query is attributed to that hop, this yields `committed_before_search` per (hop, variant). The decisive cell is `commit_no_search`: the agent committed to an answer for that hop and no search was ever attributed to it.



(a) Nemotron Nano 30B — coverage-failing under-searcher: *Truly missed* dominates (24.3% on the benchmark, 33.5% on natural), while both redundancy categories stay small.



(b) Qwen 3.5 122B — the most aggressive volume drop under natural paraphrasing (search rate 4.2× lower), with *Truly missed* expanding from 6.7% to 34.4%.

Figure 6: Extended search-calibration profiles for the remaining two models.

Result. Table 5 reports the headline rates; Figure 8 visualizes them. Across all three models, the rate of pre-search commitment rises substantially under natural framing. For Gemini, commitment-without-search jumps from 22.6% of hops on the benchmark to 51.9% on the natural rewrite. For Qwen the rise is 18.0% → 38.5%. Nemotron, which already searched the least on the benchmark, shows a smaller absolute increase.

Interpretation. The shift in commitment locus is a mechanistic explanation for the *Truly missed* growth reported in §7: under natural framing the model is not simply "forgetting" to search — it is *deciding* the answer earlier, often before any search would be triggered. This is consistent with the "context leads, parametric memory follows" phenomenon of Tao et al. (2024), re-cast at the level of the *search-decision* policy rather than the answer-generation policy.

D Missed-Hop Confounder Analysis

Motivation. *Truly missed* is the cell that grows the most under natural paraphrasing (Table 4). A potential confound is that some of these misses are not the model's failure to act on its own uncertainty but downstream casualties of an earlier mistake: if hop k resolves to a wrong entity, the model has nothing to anchor a query for hop k+1. We separate the two.

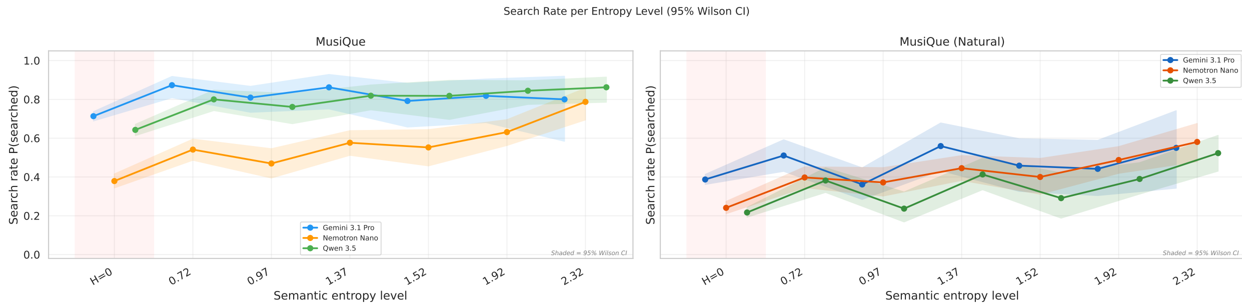


Figure 7: Search rate $P(\text{searched} \mid H = h)$ per entropy level, per model, on MuSiQue (left panel) and MuSiQue (Natural) (right panel). A well-calibrated policy would rise monotonically from H=0 to H=2.32. Observed curves are roughly flat on both datasets: behavior is approximately binary rather than graded with internal uncertainty. Shaded band = 95% Wilson CI.

Table 5: Commitment-locus rates per (model, variant). `commit_before_search`: agent committed before any search query attributed to the hop. `commit_no_search`: agent committed and no search ever attributed.

Model	Variant	Committed at all	Before search	No search at all
Gemini 3.1 Pro	Benchmark	78.5%	34.7%	22.6%
	Natural	56.1%	**52.2%**	**51.9%**
Nemotron Nano	Benchmark	52.2%	22.1%	21.5%
	Natural	54.0%	28.1%	27.8%
Qwen 3.5 122B	Benchmark	75.9%	19.6%	18.0%
	Natural	61.7%	**38.9%**	**38.5%**

Decomposition. For each hop in the *Truly missed* cell, we classify:

- **genuine_missed**: hop_index = 0, or every prior hop's parametric runs were all correct. The model could have searched and chose not to.
- **cascade_possible**: hop_index > 0 and at least one prior hop had parametric accuracy < 1.0. The model may have been "starved" of the anchor entity.

The auxiliary classifier `missed_search_answer_found_cross_hop` — used to verify that a missed gold answer was *not* present in any other hop's search results before counting the hop as missed — is the same `gpt-oss:120b` judge used for query-to-hop attribution (Appendix E.4).

Result. Table 6 and Figure 10 show that the growth in *Truly missed* under natural paraphrasing is dominated by the *genuine_missed* component for all three models. Genuine misses roughly triple for Gemini (84 → 277) and triple-plus for Qwen (105 → 369); Nemotron rises by 42% (240 → 342). This rules out the trivial explanation that the natural-paraphrase generalization gap is a downstream casualty of upstream error.

E Operational details

This appendix collects the per-stage operational decisions referenced from §3 and the results sections. None of these decisions are load-bearing for the conceptual protocol; we surface them here so that the main text can keep its focus on the behavioural claim.

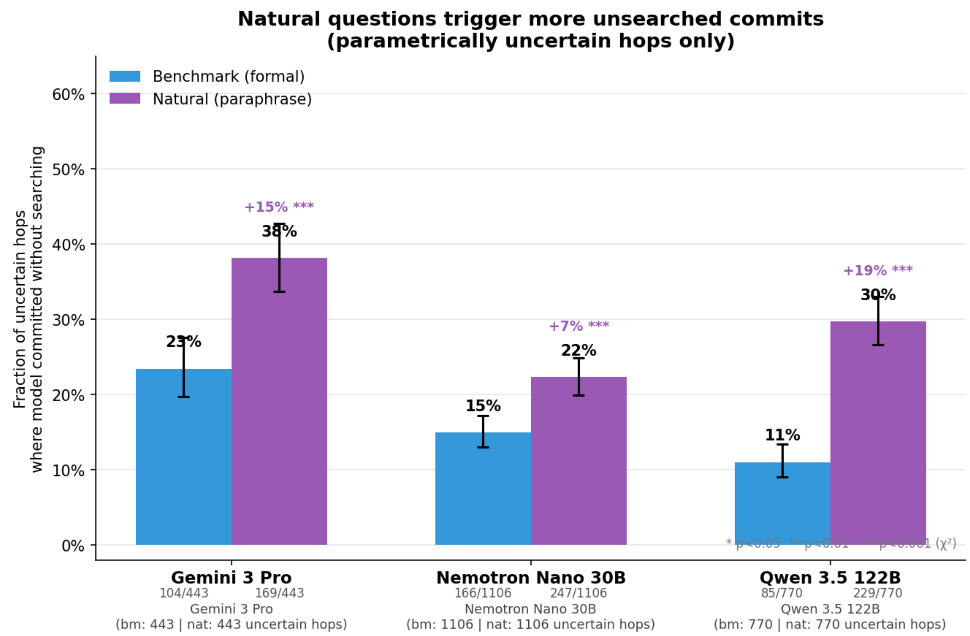


Figure 8: Pre-search commitment rate per (model, variant). Natural framing materially shifts the locus of the per-hop answer to *before* any search query is issued.

Table 6: *Truly missed* decomposition: genuine misses vs. cascade-possible misses, per (model, variant). Counts are of hops in the missed cell.

Model	**Variant**	**Cascade possible**	**Genuine missed**
Gemini 3.1 Pro	Benchmark	98	84
	Natural	254	**277**
Nemotron Nano	Benchmark	382	240
	Natural	470	**342**
Qwen 3.5 122B	Benchmark	155	105
	Natural	385	**369**

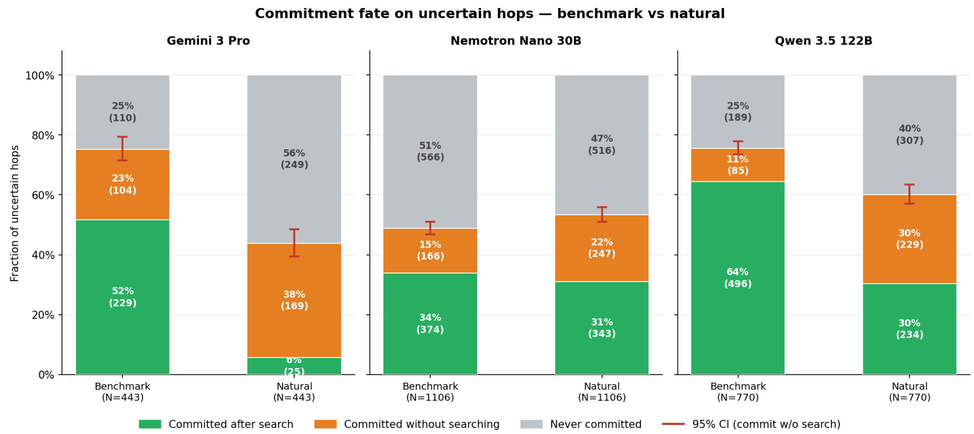


Figure 9: Commitment-fate breakdown per (model, variant). The growth of the orange "committed without searching" band under natural framing is the per-model picture of the generalization gap localized to a single behavioral mechanism.

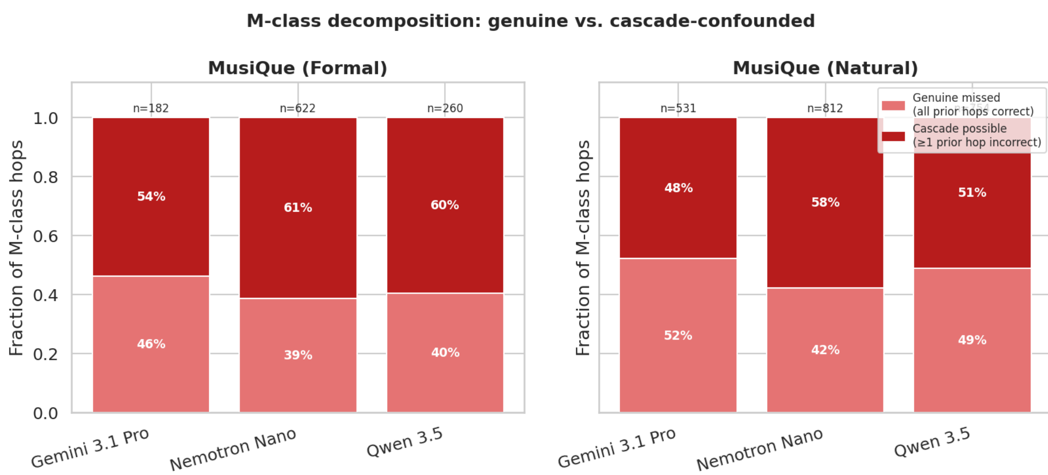


Figure 10: *Truly missed* cell decomposition. The dark (genuine_missed) component grows under natural framing for every model, ruling out cascade-driven explanations for the missed-cell expansion.

E.1 Formal protocol pseudocode

Algorithm 1 Per-hop search-calibration protocol

Require: Multi-hop question q with hops $\{h_j\}_{j=1}^{k}$ and gold answers $\{a_j\}$; model M; LLM judge J
Ensure: Six-cell behavioural label for each hop h_j

```
discard q if J flags any a_j as time-sensitive                    ▷ Stage 0
for j = 1, ..., k do                                               ▷ Stage 1: parametric uncertainty
    sample M(h_j) closed-book N=5 times; cluster by semantic equivalence via J
    E_j ← semantic entropy of the cluster distribution (in bits)
    certain_j ← [E_j = 0] ∧ [cluster ≡ a_j]
end for
T ← M(q, SEARCH)                                                   ▷ Stage 2: search-augmented trace
for search query Q_t ∈ T do                                        ▷ Stage 3: query-to-hop attribution
    ĵ_t ← J(Q_t | q, h_{1:k});   searched_{ĵ_t} ← 1
end for
for j = 1, ..., k do                                               ▷ trace-level flags
    seq_red_j ← J(earlier query in T already returned this content)
    xhop_cov_j ← J(a_j appears in results of a query with ĵ ≠ j)
end for
return six-cell label per hop from (certain_j, searched_j, seq_red_j, xhop_cov_j)
```

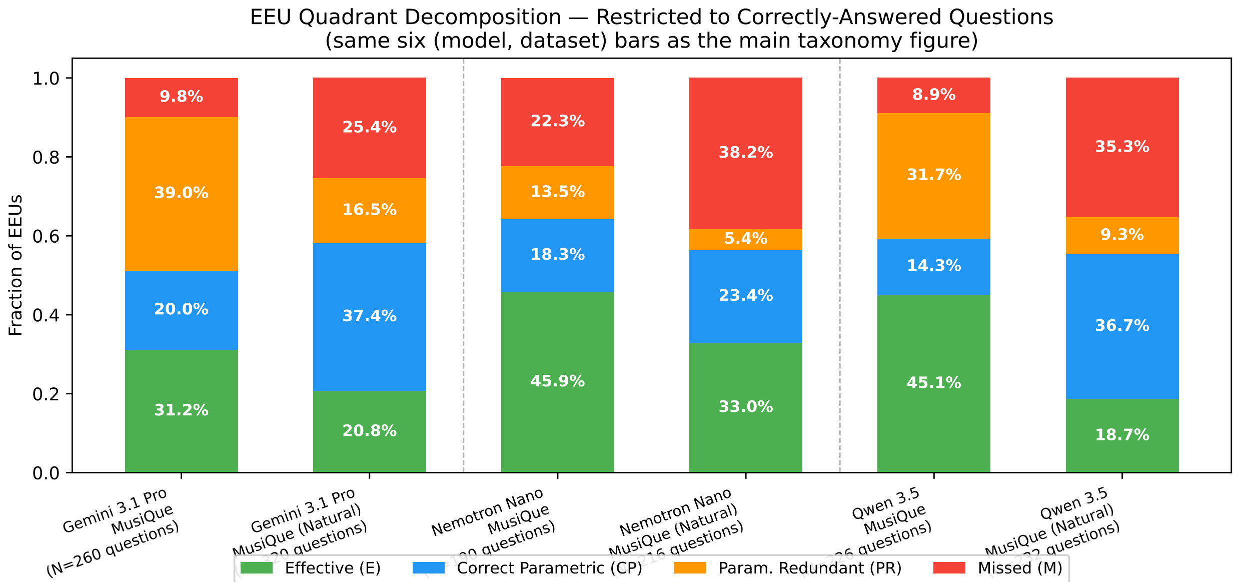


Figure 11: **Robustness check.** Reproduction of the headline taxonomy figure (Figure 4) restricted to questions on which the model produced the correct final aggregate answer. Even within this favourable sub-population, the natural-paraphrase bars show the same expansion of the red (*Missed*) segment and contraction of the green (*Effective*) segment. The calibration-and-paraphrase phenomena documented in the main text are therefore not artefacts of failure-mode questions: they appear inside the subset of questions the model actually got right.

With $N{=}5$, E_j takes one of seven discrete values $\{0, 0.722, 0.971, 1.371, 1.522, 1.922, 2.322\}$ bits. This coarse seven-value resolution is sufficient for the cross-condition contrasts in the main text but is the source of the granularity caveat in §8 (Limitations).

E.2 Cross-model statistical tests

Full test outputs (Kruskal–Wallis on entropy distributions, Mann–Whitney U on searched-hop entropies, paired Wilcoxon signed-rank on per-question search volume) are shipped with the analysis bundle in `results/consolidated_analysis/statistical_tests.csv` and summarised in Table 2 (benchmark) and Table 4 (paraphrase shift).

E.3 Models and serving

We evaluate **Gemini 3.1 Pro** (Google, proprietary, frontier-scale), **Qwen 3.5 122B** (Alibaba, open-weight, large), and **Nemotron Nano 30B** (NVIDIA, open-weight, mid-size). The selection is guided by four criteria. (i) *Vendor independence*: the three models come from entirely separate organisations with independent pre-training corpora, post-training recipes, and tool-use training data; behavioural patterns shared across all three are therefore attributable to the phenomenon rather than to a common training artifact. (ii) *Scale diversity*: the models span roughly one order of magnitude in parameter count (∼30B, 122B, frontier-scale), letting us check whether the miscalibration pattern is scale-dependent. (iii) *Access modality*: one proprietary API-only model and two open-weight models served locally, spanning the proprietary/open divide relevant for reproducibility. (iv) *Native tool-use support*: all three have been explicitly post-trained for structured tool-call generation, a prerequisite for the protocol — models that cannot issue search queries cannot be evaluated on search decisions. Non-Gemini models are served via Ollama; we do not fine-tune any model, making this a zero-shot evaluation of pretrained behavior.

E.4 Judge models

Stage 0 staleness classification uses `gpt-oss:20b`; Stage 1 semantic clustering and Stage 3 query-to-hop attribution both use `gpt-oss:120b`. Lee and Hockenmaier (2025) validate LLM judges for step-level trace evaluation, and Han et al. (2025) further show that models $\geq$70B reach or exceed human-to-human inter-rater agreement on step-level annotation tasks, justifying the use of `gpt-oss:120b` as a reliable annotator for the per-query attribution classification.

E.5 Search backend choice

We use Brave Search to reduce *provider-side confounds*: Brave operates its own independent web index and has no corporate affiliation with any of the three evaluated model vendors (Google, Alibaba, NVIDIA). While we cannot audit any vendor's training corpus and so cannot rule out incidental exposure to Brave-sourced data, the absence of a product relationship reduces the risk that a model gains an advantage from familiarity with the search backend's index structure or result format.

E.6 One-query–one-hop validation

The primary attribution maps each emitted query to a single hop. As a sanity check on this simplification, we run a second judge pass that returns the *full set* of hops each query is relevant to (prompt in Appendix F). The fraction of queries judged relevant to exactly one hop is:

Model	Benchmark MuSiQue	Natural paraphrase
Gemini 3.1 Pro	85.4%	84.4%
Nemotron Nano 30B	86.6%	81.5%
Qwen 3.5 122B	85.9%	79.2%

Across both datasets and all three models, at least 79% of queries serve a single hop; the primary attribution captures the dominant signal.

E.7 Paraphrase generation details

We use `gemma4:31b` to rewrite each of the 600 MuSiQue questions into a natural, goal-oriented paraphrase. The system prompt (Appendix F) enforces three verifiable constraints: (i) no intermediate bridge entity may be named, (ii) the "X of Y where Z did W" relative-clause pattern is forbidden, and (iii) phrasing must invite 2–4 explanatory sentences rather than a one-word answer. Semantic equivalence is guaranteed by construction: the same gold answers and the same hop decompositions from the original questions are reused in Stage 3 attribution, so any divergence in the gold answer chain would surface immediately as attribution failure. Surface-form coverage is an established strength of large instruction-tuned models for this class of constrained rewrite (Jang et al., 2026; Cheng et al., 2026). We then re-run only Stages 2–3; Stage 1 parametric uncertainty is reused from the original questions because the hop chain is semantically unchanged.

E.8 Related training-time approaches

A growing body of work tackles search-decision miscalibration at training time and is worth listing alongside S-GRPO, cited in the main text. QuCo-RAG (Min et al., 2025) injects search only when corpus statistics confirm a knowledge deficit; Unified Active Retrieval (Cheng et al., 2024b) trains a hidden-state classifier to gate retrieval on per-step uncertainty; R3-RAG (Li et al., 2025) combines outcome reward with a step-level process reward for intermediate document relevance. Our protocol provides a *diagnostic* complementary to all four methods: by measuring the calibration gap between parametric uncertainty and search decisions on a held-out set, it can quantify whether a given training method actually closes the miscalibration we document.

F Prompts

All LLM-judge calls in the protocol use the prompts reproduced below verbatim. Curly-brace placeholders are filled at runtime with the per-example fields named in the text.

F.1 Staleness Judge (Stage 0)

Model: `gpt-oss:20b` via Ollama.

```
System. You are an expert at identifying whether a factual question requires time-sensitive
knowledge.

A question is STALE if its correct answer could plausibly change over time. This includes:
- Current officeholders (presidents, ministers, CEOs, etc.)
- Current statistics (population, GDP, rankings, prices)
- Current records (tallest building, richest person, etc.)
- Ongoing relationships (current spouse, current employer, current team)
- Any fact described as "current", "present", "now", "today", or "latest"

A question is NOT STALE if its correct answer is a fixed historical fact:
- Historical events and their dates
- Births, deaths, and biographical facts about historical figures
- Geographic facts (where a place is located, what country something is in)
- Authorship, invention, or creation of things
- Past achievements and awards
- Scientific or mathematical constants

For multi-hop questions, mark as STALE if ANY hop requires time-sensitive knowledge.

User. Classify the following question as STALE or NOT_STALE.

Question: {question}

Think through whether any part of answering this question requires knowing time-sensitive
information (facts that could change over time, like current officeholders, current statistics,
or ongoing relationships).

Respond in exactly this format:
Classification: STALE or NOT_STALE
Reason: <one sentence explaining why>
```

F.2 Semantic Clusterer (Stage 1)

Model: `gpt-oss:120b` via Ollama. The structured output is a list of integer cluster IDs of length $N = 5$.

```
Question: {question}

The following {n} answers were given independently to this question:
{answers}

Group them by semantic equivalence. Two answers belong to the same cluster if they express the
same fact (minor wording differences are fine; "United States" and "USA" are the same). Use
consecutive integer cluster IDs starting from 0.
```

F.3 Paraphrase Rewriter (Paraphrase generation)

Model: `gemma4:31b` via Ollama.

```
System. You are a prompt rewriter. You will be given a multi-hop benchmark question and the
chain of sub-questions that reveal its structure. Your task is to rewrite the question so it
sounds like a genuine, curious user asking -- not like a database traversal.

Rules:
1. Do NOT mention any intermediate entity that appears only as a bridge in the hop chain. The
user asking the question would not know that entity exists.
2. Do NOT use the "X of Y where Z did W" pattern. Avoid nested relative clauses that signal the
hop structure.
3. Frame the question so the answer naturally warrants 2-4 explanatory sentences, not a single
word or number. Prefer "why", "how", "tell me about", "explain", "what can you tell me about"
phrasing where it fits naturally. If the answer is inherently factual (a location, a date), still
frame it as "I'm curious about..." or "Could you tell me a bit about..." to invite context.
```

```
4. The rewritten question must preserve the same ultimate information need and have the same gold answer.
5. Return only the rewritten question text -- no explanation, no preamble, no quotes.

User. Original benchmark question: {question}

Hop chain (do NOT expose these steps in the rewrite):
{hops}

Gold answer: {answer}

Rewrite this as a natural user question following the rules:
```

F.4 Query→Hop Attribution Judge (Stage 3)

Model: `gpt-oss:120b` via Ollama. The structured output is the integer hop index plus a confidence label.

```
You are attributing a search query issued by an AI agent to one of the sub-questions (hops) that make up a multi-hop question.

Aggregate question: {aggregate_question}

Sub-questions (hops):
{sub_questions_text}

Agent's thinking before the search query:
{thinking}

Search query: {query}

Which sub-question (hop) is this search query primarily trying to answer?
Return the 0-based hop_index (0, 1, 2, ...) or -1 if the query is about the aggregate question or doesn't clearly target a specific sub-question.
```

F.5 Commitment-Locus Judge (Appendix C)

Model: `gpt-oss:120b` via Ollama.

```
You are analyzing a transcript of an AI agent solving a multi-hop question by interleaving reasoning, search-tool calls, and search results. Each turn in the transcript is prefixed with [TURN i: ROLE ...].

Aggregate question: {aggregate_question}

Target sub-question (one hop of the chain): {hop_question}
Gold answer for this sub-question: {gold_answer}

Your task: identify the FIRST turn at which the agent commits to a specific answer for the TARGET SUB-QUESTION above (not the aggregate question and not any other hop).

"Commits" means:
- Asserts a definite answer ("X is Y", "the answer is Z"), OR
- Treats an answer as established and reuses it in subsequent reasoning (e.g. forms the next hop's search query as if the prior answer is settled).

If a turn merely lists candidate answers, expresses doubt, or fetches information without concluding, it is NOT a commitment. The very first turn that satisfies the above is the commitment turn.

Notes:
- Turn indices are 0-based and match the [TURN i: ...] prefix.
- Return -1 for first_committed_turn_index if no commitment occurs anywhere in the transcript.
- matches_gold compares the COMMITTED answer to the GOLD answer; commitment exists even when wrong.

==== TRANSCRIPT ====
{transcript}
==== END TRANSCRIPT ====
```